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Inventor(s)	Gill; Davinder

Coronavirus vaccine and methods of use thereof

Abstract

An immunogenic composition effective for eliciting an immune response against cells that present coronavirus S protein and/or coronavirus M protein derived antigens on a virus-like-particle (VLP) system. In a method embodiment, the antigen presenting VLP is administered to a mammal, such as a human, to elicit an immune response against coronavirus S protein and/or coronavirus M protein. A preferred method embodiment may include at least one additional dose of immunogenic composition to enhance the immune response effectiveness of the coronavirus vaccine.

Inventors:	Gill; Davinder (Andover, MA)
Applicant:	Gill; Davinder (Andover, MA)
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USPC: None

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Primary Examiner: Foley; Shanon A.

Attorney, Agent or Firm: WCF IP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims priority to U.S. Provisional Application 63/037,367 filed on Jun. 10, 2020. The complete content thereof is herein incorporated by reference.

SEQUENCE LISTING

(1) This document incorporates by reference an electronic sequence listing text file, which was

electronically submitted along with this document. The text file is named 15660001AA_ST25, is 53 kilobytes, and was created on Jun. 7, 2021.

FIELD OF THE INVENTION

(2) The invention generally relates to a coronavirus vaccine and particularly to an immunogenic composition comprising polypeptide antigens that are derived from conserved C-terminal domains of coronavirus spike protein (S protein). The composition may further comprise at least one polypeptide antigen derived from the first Heptad Repeat (HR1) and/or the second Heptad Repeat (HR2) domain of S protein and/or coronavirus membrane protein (M protein) and/or coronavirus receptor binding domain (RBD). The present invention also contemplates a vaccine composition comprising coronavirus antigens presented on a virus-like-particle (VLP). In addition, the invention relates to a method of inducing an immune response to coronavirus antigens to protect a subject from acquiring COVID-19.

BACKGROUND

(3) The COVID-19 pandemic caused by severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) has caused untold devastation worldwide. Similar to other human coronaviruses, such as the Middle East respiratory syndrome coronavirus (MERS-CoV) and SARS-CoV-1, SARS-CoV-2 may have a zoonotic origin. To date, 160 million people have been infected and 3.3 million have died worldwide, with mortality and morbidity still rising in various parts of the world. Apart from the medical and public health havoc, the SARS-CoV-2 pandemic has also caused unprecedented social, economic and governmental upheaval unseen in modern times.

(4) As a result, there has been a thrust towards rapid development of vaccines and therapeutics to tackle this urgent problem. The prophylactic vaccines are based on diverse platforms that include RNA technology, non-replicating viral vectors, inactivated virus, subunit vaccines, as well as DNA vaccines. A number of therapeutic approaches using monoclonal antibodies have also been reported. However, there are limited pre-clinical data published on the SARS-CoV-2 vaccines under development. Long range safety studies have not yet been reported with test vaccines against SARS-CoV-2. In the case of therapeutic antibodies, based on published research, focus has primarily been on isolation techniques, structural models and mechanism of action of therapeutic antibodies against SARS-CoV-2. Further investigation on the safety aspects of either vaccines or therapeutic antibodies against SARS-CoV-2 is needed.

(5) SARS-CoV-2 pathophysiology shows that the COVID-19 disease clinically manifests in multiple forms. Mild to moderate disease has been reported in 80% of cases, severe disease in ~15% cases and critical disease in ~5% cases. Subjects with mild disease recover with little to no medical intervention while severe and critical disease require hospitalization and, in some cases, intensive care. Some patients may suffer from life-threatening acute respiratory distress syndrome (ARDS) with possible fatal outcomes. Severe disease is seen more often in males, particularly in those with co-morbidities such as cardiovascular disease and diabetes. Paradoxically, patients with severe disease have increased IgG response and higher titers of total antibodies against SARS-CoV-2, which are associated with worse outcomes. As such, the role of immune cells such as macrophages, dendritic cells and T cells in COVID-19 disease has not been systematically studied. Antibody dependent enhancement (ADE) of SARS-CoV-2 may result in potential side effects in those patients who produce undesirable non-neutralizing antibodies against the virus.

(6) SARS-CoV-2 infects cells through the ACE2 receptor expressed on type II pneumocytes in the lung. However, ACE2 is also expressed in the heart, kidney, intestine as well as the vascular endothelium. The major determinant of the ACE2 cognate binding is the homotrimeric spike glycoprotein (S) expressed as 70-80 copies on the surface of SARS-CoV-2. The S protein consists of two domains—S1 and S2—that are linked by a fusion peptide FP. S1 domain contains the ACE2 receptor binding domain (RBD) while the S2 domain promotes fusion of the virus into the cell membrane using a polybasic furin cleavage site located within the S1-S2 boundary.

(7) Majority of the vaccine candidates currently being clinically tested make use of the S protein of

SARS-CoV-2 alone as the main immunogen, mostly in a non-specific manner. In the case of non-replicating virus vectors, the S protein is cloned and expressed in adenovirus vectors. Similarly, for RNA based vaccines, the S protein mRNA is encapsulated in lipid nanoparticles and used for intramuscular injections. The DNA based vaccine also encodes the S protein and is electroporated into subjects for vaccinations. Thus, while initial attempts at developing vaccines and therapeutic antibodies against SARS-CoV-2 focused solely on the use of S protein, it is increasingly becoming clear that without rational considerations into immunogen design, formulation and manufacturer, clinical trials could get entrapped in safety pitfalls, thereby jeopardizing global efforts at controlling the pandemic.

(8) Three major areas of concerns regarding safety of current vaccines against SARS-CoV-2 have emerged. One of the concerns of the currently developed vaccines is that some of the vaccines may trigger antibody dependent enhancement (ADE) of viral infection. ADE is a phenomenon wherein antibodies that are elicited against a viral antigen actually enhance the uptake of the virus, causing more severe disease. The enhancement is typically the result of non-neutralizing or sub-neutralizing antibodies generated against viral antigen (Iwasaki et. al., 2020). While the role of ADE in SARS-CoV-2 has not been fully established, several lines of evidence suggest that it could potentially be of significant safety concern in the clinic (Kamikubo et. al., 2020).

(9) A second challenge to developing a safe vaccine is the need to reduce an off-target pathological activation of the immune cells. There is emerging evidence that TH17 responses can direct certain cellular responses upon vaccination with inactivated vaccines and those based on viral vectors, and that the TH17 activation leads to up-regulation of the pro-inflammatory cytokines IL-6 and IL-1 β (Blanco-Melo et. al., 2020, Liu et. al., 2020). IL-6 upregulation has been prominently observed in patients with severe COVID-19 disease (Herold et. al., 2020). In rhesus macaques vaccinated with a modified vaccinia Ankara (MVA) virus encoding the SARS-CoV-1 S glycoprotein, anti-spike antibodies promoted MCP1 and IL-8 production in alveolar macrophages causing acute lung injury (Liu et. al., 2019).

(10) An exemplary mouse study showed that, when animals were challenged with SARS-CoV-1 after immunization with either an inactivated viral vaccine or rDNA-based vaccine with or without alum, the Th2-type pathology with prominent eosinophil infiltration with eosinophil scores were significantly lower for non-vaccine groups than for vaccine groups of across the tested mouse strains (Tseng et. al., 2012). Spleen atrophy and lymph node necrosis have also been reported to be decreased in COVID-19 patients suggesting immune mediated pathology in SARS-CoV-2 infections (Feng et. al., 2020).

(11) A third concern regarding of some vaccines against SARS-CoV-2 is a possibility of triggering endothelium inflammation. Endothelial cells express ACE2, the receptor bound by SARS-CoV-2 to infect lung epithelial cells. Not surprisingly, in addition to pulmonary complications, additional clinical symptoms of COVID-19 include high blood pressure, thrombosis and pulmonary embolism. This raises the question of whether the endothelium is a key target organ of SARS-CoV-2 (Sardu et. al., 2020). Of note, supernatants from SARS-CoV-2 infected capillary organoid cultures could also infect Vero cells demonstrating that the production of viable progeny virus. Postmortem analysis of a SARS-CoV-2 patient with hypertension showed evidence of direct viral infection of the endothelial cell and diffused endothelial inflammation causing vascular dysfunction by shifting the equilibrium towards vasoconstriction, ischemia and a procoagulant state (Fox et. al., 2020; Tian et. al., 2020; Varga et. al., 2020).

(12) Therapies such as afucosylated or defucosylated monoclonal antibodies have been proposed, however, in some instances, such therapies are pathogenic.

(13) Additionally, there is a great need to quickly develop and verify efficacy and safety of a vaccine for the new emergent SARS-CoV-2 virus variants, and subsequently manufacture the vaccine on a very large scale, to meet immediate population demands. Use of shorter antigenic peptides that are developed from computer-based rational designs provide several advantages in

comparison to conventional vaccines made of dead or attenuated pathogens or inactivated toxins. This version of polypeptide antigens may be synthesized rapidly and produced with much lower cost.

(14) Thus, there is a need in the art for an improved SARS-CoV-2 vaccine with effective prophylactic properties and ability to control potential side effects to the lowest degree or to none. Further, there is a need for diversifying new SARS-CoV-2 vaccine targets and more specifically to new targets based on the conserved regions for being effective against other potential variants, in the most cost effective and safety-oriented manner.

SUMMARY OF THE INVENTION

(15) The disclosure relates to the field of viral vaccines and methods of use of such vaccines to protect a subject from virus infection. In particular, the invention is an immunogenic composition comprising at least one antigenic polypeptide and methods of use to evoke an immune response to one or more coronavirus spike (S) protein and/or membrane (M) protein to protect an immunized subject from acquiring a viral-related disease such as COVID-19. The antigenic polypeptides or fragments derived from the S protein and/or M protein may be included in the composition separately or fused by a linker region. The antigens are presented on the surface of Virus-Like Particles (VLP) for inhibiting the fusion of coronavirus particles during the viral entry to a host cell. One of the main advantageous features of the present invention is to selectively target the viral fusion stage by introducing at least one antigenic polypeptide or fragment derived from at least one conserved domain of the S protein from SARS-CoV-2 or any other coronaviruses with the conserved domain, such as the first heptad repeat (HR1) and/or the second heptad repeat (HR2), in which the at least one antigenic polypeptide is presented on the surface of VLP as a path to develop a safe and affordable vaccine. By presenting rationally designed antigens on VLPs, host immune responses that block viral fusion will result in prevention of unwanted cytokine production and immunopathology. The S protein derived antigens may be manufactured in eukaryotic cells (e.g., mammalian cells, plant cells, fungal cells, etc.), more preferably in mammalian cells, to ensure proper folding and glycosylation of viral proteins as in the human system.

(16) One aspect of the invention is a vaccine composition that is able to induce an immune response against a coronavirus, comprising a VLP presenting at least one polypeptide or fragment of the coronavirus as an immunogen, wherein the at least one antigenic polypeptide or fragment may be derived from HR1 domain and/or HR2 domain of S protein and/or M protein from the coronavirus. In some embodiments, the at least one antigenic polypeptide or fragment is derived from both HR1 and HR2 domains and/or the linker regions between the HR1 and HR2 domains. In other embodiments, the at least one antigenic polypeptide or fragment may be derived from N-terminal or C-terminal region of the coronavirus M protein.

(17) Another aspect of the invention is a vaccine composition that is able to induce an immune response to coronavirus and to inhibit viral fusion, wherein the at least one antigenic polypeptide or fragment presented on VLP has a sequence selected from the group consisting of SEQ ID NOs:1 to 166. In some embodiments, the VLP has a sequence as set forth in SEQ ID NO: 167 or SEQ ID NO: 168. In some embodiments, the at least one antigenic polypeptide includes a plurality of polypeptides as set forth in SEQ ID Nos 1-7, 83-102, and 135-150

(18) Another aspect of the invention is a method of inducing an immune response to at least one coronavirus antigen in a subject in need thereof, comprising the step of administering an immunogenic composition comprising a VLP presenting at least one antigenic polypeptide or fragment derived from S protein and/or M protein to the subject. In some embodiments, the method further comprises allowing a suitable period of time to elapse and administering at least one additional dose of the immunogenic composition. The immunogenic composition comprising the vaccine can be administered intramuscularly or intradermally with a hypodermic, transdermic or intradermal needle or with a needle-free device. In some embodiments, the immunogenic composition comprising the vaccine can be administered by intranasal and/or ocular delivery. In

some embodiments, at least one adjuvant may be included in the vaccine composition. In other embodiments, 1-150 μ g of the antigen presenting VLP is administered to the subject in each dose. In addition, a suitable period of time is defined as a time sufficient for producing antibodies against the immunogenic composition of the present invention in a subject.

(19) Other features and advantages of the present invention will be set forth in the description of invention that follows, and in part will be apparent from the description or may be learned by practice of the invention. The invention will be realized and attained by the compositions and methods particularly pointed out in the written description and claims hereof.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate embodiments of the invention and, together with a general description of the invention given above and the detailed description given below serve to explain the invention.

(2) FIG. 1 shows a structural organization of various domains within SARS-CoV-2 S protein. SS, signal sequence; S2', S2' protease cleavage site; FP, fusion peptide; HR1, heptad repeat 1; CH, central helix CD, connector domain; HR2, heptad repeat 2; TM, transmembrane domain; CT, cytoplasmic tail. The S protein also contains a predicted set of 24 N-glycans and 4 O-glycan for each monomer of which 17 N-glycans have been confirmed using cryo-EM.

(3) FIG. 2 shows a sequence variability within N-terminal domains of SARS-CoV-2 S protein.

(4) FIG. 3A-B show a front view (A) and a top view (B) of sequence conservation within C-terminal domains of SARS-CoV-2 S protein.

(5) FIG. 4A-B show a 6-helix bundle fusion complex formed between HR1 and HR2, front view (A) and top view (B). The long HR1 helices are packed in the interior of the bundle whereas the short HR2 helices are located on the exterior.

(6) FIG. 5A-D show a composition of the proposed antigens SC2SHR1 (A), SC2HSHRL (B), SC2HSHR2 (C) and SC2HSHR2L (D) in the cases where high-resolution structures of S or M protein are available. 6VXX.PDB, 6LXT.PDB and 6M3W.PDB are shown in the figure.

DETAILED DESCRIPTION

(7) Embodiments of the invention relate to an immunogenic composition comprising at least one antigenic polypeptide or fragment thereof, wherein the at least one polypeptide or fragment antigen thereof is derived from coronavirus spike (S) protein and/or membrane (M) protein. In particular, HR1 domain and/or HR2 domain of SARS-CoV-2 S protein is contemplated as the protein in which at least one antigenic polypeptide or fragment thereof is derived from. When incorporated together, the polypeptide or fragment derived from HR1 and HR2 may be fused, chemical and/or bio-conjugated to each other by a linker region and presented on the surface of Virus-Like-Particles (VLP) for inhibiting the fusion of SARS-CoV-2 virus during the viral entry to a host cell. The disclosure also relates to a method of use of such immunogenic composition to protect a subject from COVID-19 infection.

(8) The vast majority of the vaccines that are currently being developed target the spike (S) protein of SARS-CoV-2 in a non-specific manner, opening the possibilities of unwanted immune responses and pathologies clinically reported in severe and critical cases of COVID-19. In particular, acute lung injury, immunopathology and endothelial dysfunction have been described in deceased patients, raising questions about the fundamental mechanism of COVID-19 etiology. In the present disclosure, an alternate strategy based on rational design that selectively targets blockade of viral fusion as a path to develop a safe and affordable vaccine against SARS-CoV-2 is described. The present invention displays antigenic polypeptides or fragments that are derived from specific, membrane-proximal, conserved domains within the SARS-CoV-2 S protein as well as from

membrane glycoprotein as immunogens. By presenting rationally designed antigens on VLP, host immune responses that block viral fusion will result in prevention of unwanted cytokine production and immunopathology. Manufacture in mammalian cells ensures proper folding and glycosylation of viral proteins as in the human system, further strengthening vaccine safety. Combined with proven, scalable, cost-effective manufacturing technologies, the present invention may provide a shelf life and lower cost of goods for people living in low-income countries.

(9) In preferred embodiments, a vaccine composition of the present invention induces an immune response to coronavirus and inhibits viral fusion. The immunogenic composition comprises at least one antigenic polypeptide or fragment thereof. In preferred embodiments, one or more antigens are presented on VLP. In other embodiments the antigens may be displayed or included in a VLP by a plurality of methods described below. In some embodiments, the antigenic polypeptide or fragment has 80% or more, preferably 90% or more, more preferably 95% or more sequence identity with a sequence selected from the group consisting of SEQ ID NOs:1 to 166. In some embodiments, the VLP has 80% or more, preferably 90% or more, more preferably 95% or more sequence identity with a sequence selected from the group consisting of SEQ ID NOs:167-168.

(10) For each cDNA sequence presented herein, the invention includes the mRNA equivalent of the cDNA, meaning that the invention includes each cDNA sequence wherein each T is replaced by U. Exemplary antigenic peptide sequences include:

(11) Amino Acids 1-31 Covering Amino Terminus of M Protein:

(12) TABLE-US-00001 (SEQ ID NO: 151)
MADSNGTITVEELKKLLEQWNLVIGFLFLTW

Corresponding DNA Sequence:

(13) TABLE-US-00002 (SEQ ID NO: 152)
ATGGCTGATTCTAATGGGACAATTACAGTCGAAGAGCTTAAGAAATC
GCTGGAGCAATGGAATCTTGTTATTGGCTTTCTCTTCCTGACCTGG

Amino Acids 130-160 from Carboxy Terminus of M Protein:

(14) TABLE-US-00003 (SEQ ID NO: 153) TRPLLESELVIGAVILRGHLRIAGHHLGRCD
Corresponding DNA Sequence:

(15) TABLE-US-00004 (SEQ ID NO: 154)
ACCCGACCCCTCCTTGAATCCGAACCTTGTTATTGGCGCTGTCATTCT
CCGCGGACACCTTAGAATTGCTGGACATCACCTTGGACGCTGTGAT

Amino Acids 913-984 Covering the First Heptad Repeat HR1 of S Protein:

(16) TABLE-US-00005 (SEQ ID NO: 155)
QNVLYENQKLIANQFNSAIGKIQDSLSTASALGKL
QDVVNQNAQALNTLVKQLSSNFGAISSVLNDILSRL

Corresponding DNA Sequence:

(17) TABLE-US-00006 (SEQ ID NO: 156) CAGAATGTCCTTTACGAAAATCAGAAA
CTCATCGCAAATCAATTCAACTCAGCAATCGGAAAAATTCAAGATTCCCTCTCT
TCTACCGCATCTGCTCTTGGAAGCTTCAAGATGTCGTCAATCAAAATGCCCAA
GCTCTCAACACTCTCGTTAAACAACCTCTCTTCTAACTTTGGTGCCATATCCTCC
GTCCTCAATGATATCCTTTCCCGCCTG

Amino Acids 1147-1170 Covering the Linker Region Connecting HR1 and HR2:

(18) TABLE-US-00007 (SEQ ID NO: 157) SFKEELDKYFKNHTSPDVDLGDIS
Corresponding DNA Sequence:

(19) TABLE-US-00008 (SEQ ID NO: 158) AGTTTCAAAGAAGAACTTGACAAATACT
TTAAAAATCATACCTCCCCTGACGTGGACCTTGGCGATATCTCC

Amino Acids 1171-1212 Covering the Second Heptad Repeat HR2 of S Protein:

(20) TABLE-US-00009 (SEQ ID NO: 159)
GINASVVNIQKEIDRLNEVAKNLNESLIDLQELGKYEQYIKW

Corresponding DNA Sequence:

(21) TABLE-US-00010 (SEQ ID NO: 160) GGAATTAACGCATCCGTGGTTAACATA
CAGAAAGAAATTGACAGACTGAACGAA GTGGCCAAGAACCTTAACGAATCTCTC
ATAGACCTTCAAGAGCTGGGAAAGTAC GAACAATACATAAAATGG

Amino acids 1147-1212 covering both the linker as well as HR2 regions:

(22) TABLE-US-00011 (SEQ ID NO: 161)
SFKEELDKYFKNHTSPDVDLGDLSGINASVVNIQ
KEIDRLNEVAKNLNESLIDLQELGKYEQYIKW

Corresponding DNA Sequence:

(23) TABLE-US-00012 (SEQ ID NO: 162)
AGTTTTTAAGGAAGAAGAACTGGACAAATATTTCAAGAATCATA
CATCTCCAGACGTGGACCTGGGCGACATTTCTGGCATTAA
CGCATCCGTGGTTAACATTCAAAAAGAAATTGATAGACTG
AACGAAGTGGCTAAAAATCTGAACGAGTCCCTTATCGATC
TGCAGGAATTGGGAAAATACGAGCAATACATCAAATGG

Amino Acids 318-541 Representing RBD of S Protein:

(24) TABLE-US-00013 (SEQ ID NO: 163)
GFRVQPTESIVRFPNITNLCPPGEVFNATRFASVYAWNR
KRISNCVADYSVLYNSASFSTFKCYGVSPTKLNDLCFTN
VYADSFVIRGDEVQRQIAPGQTGKIADYNYKLPDDFTGCV
IAWNSNNLDSKVGGNYNLYRLFRKSNLKPFERDISTEI
YQAGSTPCNGVEGFNCYFPLQSYGFQPTNGVGYQPYRVV
VLSFELLHAPATVCGPKKSTNLVKNFCCVNFNFNGLTGT
GVLTESNKKFLPFQQFGRDIADTTDAVRDPQT

Corresponding DNA Sequence:

(25) TABLE-US-00014 (SEQ ID NO: 164)
GGCTTCAGAGTCCAACCAACAGAGTCCATCGTGAGGTTC
CCCAATATTACTAATCTGTGCCCCCTTTGGTGAGGTGTTT
AATGCTACCAGATTTGCCTCTGTCTATGCATGGAATCGG
AAGCGGATTAGTAACTGCGTCGCCGACTATAGTGTTCTC
TATAATTCCGCTAGTTTCTCTACGTTCAAATGCTATGGC
GTCTCCCCGACAAAGCTCAATGACTTGTGTTTCACTAAC
GTCTACGCTGATTCTTTCGTGATCCGCGGTGATGAAGTG
CGCCAGATCGCCCCAGGACAAACCGGAAAAATCGCTGAT
TACAATTACAACTCCCCGACGACTTCACCGGCTGCGTT
ATTGCCTGGAAGTCTAACAATCTGGACAGCAAGGTTGGC
GGCAATTATAACTATCTGTACCGCCTGTTTCGGAAGTCA
AATCTCAAACCATTCGAACGCGATATTAGTACAGAAATC
TATCAGGCTGGCAGCACCCCCTGTAAACGGTGTTGAAGGG
TTTAATTGTTATTTCCCTCTCCAATCATACGGTTTCCAG
CCCACAAACGGCGTTGGGTACCAACCATAACCGAGTCGTT
GTTCTGTCTTTTGAAGTTCTCCATGCTCCAGCTACTGTT
TGTGGACCGAAGAAGAGCACCAATCTTGTCAAAAATAAA
TGCGTGAATTTTAACTTCAATGGTCTTACAGGTACCGGC
GTGCTTACCGAAAGTAACAAAAAATTTCTCCCTTTTCAG
CAGTTCGGACGAGACATTGCAGATACCACCGACGCCGTG AGAGATCCACAGACC

Amino Acids 318-541 Representing RBD of S Protein with K417N/E484K/N501Y Mutations:

(26) TABLE-US-00015 (SEQ ID NO: 165)
GFRVQPTESIVRFPNITNLCPPGEVFNATRFASVYAWNR
KRISNCVADYSVLYNSASFSTFKCYGVSPTKLNDLCFTN
VYADSFVIRGDEVQRQIAPGQTGNIADYNYKLPDDFTGCV

IAWNSTNLSKYNLYRKLFRKSNLKPFERDISTEI
YQAGSTPCNGVKGFNCYFPLQSYGFQPTYGVGYQPYRVV
VLSFELLHAPATVCGPKKSTNLVKNKCVNFNFNGLTGTG
VLTESNKKFLPFQQFGRDIADTTDAVRDPQT

Corresponding DNA Sequence:

(27) TABLE-US-00016 (SEQ ID NO: 166)

GGCTTCAGAGTCCAACCAACAGAGTCCATCGTGAGGTTC
CCCAATATTACTAATCTGTGCCCCCTTTGGTGAGGTGTTT
AATGCTACCAGATTTGCCTCTGTCTATGCATGGAATCGG
AAGCGGATTAGTAACTGCGTCGCCGACTATAGTGTTCTC
TATAATTCCGCTAGTTTCTCTACGTTCAAATGCTATGGC
GTCTCCCCGACAAAGCTCAATGACTTGTGTTTCACTAAC
GTCTACGCTGATTCTTTCGTGATCCGCGGTGATGAAGTG
CGCCAGATCGCCCCAGGACAAACCGGAAATATCGCTGAT
TACAATTACAACTCCCCGACGACTTCACCGGCTGCGTT
ATTGCCTGGAACCTCTAACAATCTGGACAGCAAGGTTGGC
GGCAATTATAACTATCTGTACCGCCTGTTTCGGAAGTCA
AATCTCAAACCATTCGAACGCGATATTAGTACAGAAATC
TATCAGGCTGGCAGCACCCCCTGTAAACGGTGTTAAAGGG
TTTAATTGTTATTTCCCTCTCCAATCATACGGTTTCCAG
CCCACATACGGCGTTGGGTACCAACCATAACCGAGTCGTT
GTTCTGTCTTTTGAACCTTCTCCATGCTCCAGCTACTGTT
TGTGGACCGAAGAAGAGCACCAATCTTGTCAAAAATAAA
TGCGTGAATTTTAACTTCAATGGTCTTACAGGTACCGGC
GTGCTTACCGAAAGTAACAAAAAATTTCTCCCTTTTCAG
CAGTTCGGACGAGACATTGCAGATACCACCGACGCCGTG AGAGATCCACAGACC

Exemplary VLP Amino Acid Sequence:

(28) TABLE-US-00017 (SEQ ID NO: 167)

GGGGGGMENITSGFLGPLLVLQAGFLLTRILTIPQSLDSW
WTSLNFLGGSPVCLGQNSQSPTSNSHPTSCPPICPGYRWMC
RRRFIIFLILLCLIFLLVLLDYQGMLPVCPLIPGSTTTS
TGPKKTCTTPAQNSMFPSCCCTKPTDGNCTCIPISSWAF
AKYLWEWASVRFSWVSLLPVFVQWFVGLSPTVWLSAIWMMW
YWGPSLYSIVSPFIPLLPIFLCLWVYI

Exemplary VLP DNA Sequence:

(29) TABLE-US-00018 (SEQ ID NO: 168)

GGCGGCGGCGGCGGAGGCATGGAAAATATCACTTCTGGATT
TCTGGGGCCCCTCCTGGTTCTGCAGGCAGGCTTTTTCCTTT
TGACACGCATCCTGACTATCCCACAATCCCTTGACTCATGG
TGGACATCACTGAACTTCCTTGGCGGTTCTCCCGTTTGCCT
TGGCCAGAATTCCCAGTCACCCACTTCTAATCATTCTCCCA
CATCTTGCCCTCCTATCTGCCCAGGCTACCGATGGATGTGC
AGAAGACGCTTCATTATCTTCCTGTTCATTTTGCTGCTGTG
TCTGATCTTTCTCTTGGTCTTGCTTGATTATCAAGGCATGT
TGCCCGTGTGTCCCCTCATTCCAGGATCAACAACGACTTCC
ACAGGCCCCCTGCAAAACGTGCACCACACCAGCCCAAGGAAA
TAGCATGTTCCCCTCTTGCTGTTGCACTAAACCTACGGACG
GCAACTGTACCTGTATCCCGATAACCCTCTTCTTGGGCTTTT
GCTAAATATCTCTGGGAATGGGCTTCCGTCAGATTCTCTTG
GGTGAGCCTTCTTGTCCCCTTCGTGCAATGGTTCGTTGGAC

TCAGTCTGCTTGGCTCAGCGCAATCTGGATGTGG
TACTGGGGACCATCTCTCTACAGCATTGTTTCACCCTTTAT
CCCCCTGCTTCCAATCTTGTCTGCTTGTGGGTTTATATAT AAACGCGT

(30) Coronaviruses constitute the subfamily Orthocoronavirinae, in the family Coronaviridae, order Nidovirales, and realm Riboviria. Coronaviruses have four genera: alpha-, beta-, gamma-, and delta-coronaviruses. They are enveloped viruses with a positive-sense single-stranded RNA genome and a nucleocapsid of helical symmetry. The genome size of coronaviruses ranges from approximately 26 to 32 kilobases. Exemplary coronaviruses that may be treated with the compositions of the disclosure include, but are not limited to, SARS-Cov, SARS-Cov-2, MERS-Cov, HCoV-OC43, HCoV-HKU1, HCoV-229E, and HCoV-NL63.

(31) As used herein, the term “severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2)”, “2019 novel coronavirus (2019-nCoV)”, “human coronavirus 2019 (HCoV-19)” or “severe acute respiratory syndrome-related coronavirus (SARSr-CoV)” refers to virus comprising a virion with 50-200 nanometers in diameter and a genomic size of about 30 kilobases, encoding multiple structural proteins, such as the S (spike), E (envelope), M (membrane) and N (nucleocapsid), and non-structural proteins. Coronaviruses are a group of related RNA viruses that cause diseases in mammals and birds. In humans and birds, they cause respiratory tract infections that can range from mild to lethal. Mild illnesses in humans include some cases of the common cold (which is also caused by other viruses, predominantly rhinoviruses), while more lethal varieties can cause SARS, MERS, and COVID-19.

(32) Since SARS-CoV-2 shares 80% sequence homology with SARS-CoV-1, the antibodies against S protein in SARS-CoV-2 may also trigger immune response against SARS-Cov-1. Thus, other virus types such as SARS-CoV (i.e., SARS-CoV-1) and MERS-CoV that are similar in virion structure may also be subjected to the present invention. As used herein, the term “S protein” or “Spike protein” is used to refer to a knoblike structured (i.e., spikes) peplomer, which is composed of glycoprotein to project from the lipid bilayer of the surface envelope of an enveloped virus. The “spike protein” or “S protein” is interchangeably referred to a protein and/or a glycoprotein. Furthermore, the sequences encoding the SARS-CoV-2 glycoprotein may also be referred to as a peptide or amino acid sequence.

(33) As used herein, the terms “polypeptide”, “short protein”, “fragment of protein”, “polypeptide or fragment”, “antigenic polypeptide or fragment” and “peptide” are used interchangeably and refer to chains of amino acids comprising between 2, or 3, or 4, or 5, or 6, or 7, or 8, or 9, or 10, or 11, or 12, or 13, or 14, or 15 and 10, or 11, or 12, or 13, or 14, or 15, or 20, or 25, or 30, or 35, or 40, or 45, or 50 or 55 or 60 or 100 or 150 or 200 or 300 or 350 or 400 or 500 amino acids. As used herein, the term “epitope” or “T cell epitope” refers to a sequence of contiguous amino acids contained within a protein antigen that possess a binding affinity to a T cell receptor when presented on the surface of antigen presenting cells. An epitope is antigen-specific but not individual specific. An epitope, a T cell epitope, a polypeptide, a fragment of a polypeptide or a composition comprising a polypeptide or a fragment thereof is “immunogenic” for a specific human individual if it is capable of inducing an immune cell response in that individual. In some embodiments, an “immune response”, “T cell response” or “immunogenic response” are used interchangeably and may further include an antibody response. As used historically, the term “antigen” is used to designate an entity that is bound by an antigen-specific antibody or B-cell antigen receptor.

(34) As used herein, the term “antigens”, “proteins”, “peptides”, “polypeptides”, “fragments”, or “epitopes” may be used interchangeably. In particular, an “antigen” refers to a molecule containing one or more epitopes (either linear, conformational or both) that will stimulate a host's immune-system to make a humoral and/or cellular antigen-specific response. The term is used interchangeably with the term “immunogen”. Further, antigenic polypeptide or fragment “derived from” a particular viral protein or protein domain refers to a full-length or near full-length viral protein or domain, as well as a fragment thereof, or a viral protein with internal deletions.

Accordingly, the polypeptide may comprise the full-length sequence, fragments, truncated and partial sequences, as well as analogs and precursor forms of the reference molecule. In addition, the term “derived” may refer to construction of a peptide based on the knowledge of a representative protein domain sequence using any one of several suitable means, including, by way of example, isolation or synthesis. Thus, the term includes variations of the specified polypeptide.

(35) In some embodiments, the antigenic peptides as described herein are 5 to 150 residues in length, e.g. 10 to 100 residues. The antigenic peptides may include consecutive or nonconsecutive sequences from the coronavirus viral domains.

(36) The antigenic peptides described herein may comprise epitopes, i.e. amino acids that bind to an antibody generated in response to such sequence. An epitope for use in the subject invention is not limited to a polypeptide having the exact sequence of the portion of the parent protein from which it is derived. Indeed, viral genomes are in a state of constant flux and contain several variable domains which exhibit relatively high degrees of variability between isolates. Thus, the term “epitope” encompasses sequences identical to the native sequence, as well as modifications to the native sequence, such as deletions, additions, and substitutions (generally conservative in nature).

(37) As used herein, the term “conformational epitope” refers to a recombinant epitope having structural features native to the amino acid sequence encoding the epitope within the full-length natural protein. Native structural features include, but are not limited to, glycosylation and three-dimensional structure. The length of the epitope-defining sequence can be subject to wide variations as these epitopes are believed to be formed by the three-dimensional shape of the antigen (e.g., folding). Thus, amino acids defining the epitope can be relatively few in number, but widely dispersed along the length of the molecule (or even on different molecules in the case of dimers, etc.), being brought into correct epitope conformation via folding. The portions of the antigen between the residues defining the epitope may not be critical to the conformational structure of the epitope. For example, deletion or substitution of these intervening sequences may not affect the conformational epitope provided sequences critical to epitope conformation are maintained (e.g., cysteines involved in disulfide bonding, glycosylation sites, etc.).

(38) As used herein, the term “Virus-Like Particle (VLP)” refers to molecules that closely resembles viruses but are lacking viral genetic materials. Use of VLPs derived from the Hepatitis B virus (HBV) and composed of the small HBV derived surface antigen (HBsAg) are well known in the art. A plurality of detailed VLP formation and release methods are described in U.S. Pat. Nos. 7,951,384 and 9,352,031, herein incorporated by reference. In some embodiments, the present invention includes VLP presenting S protein derived antigens on its surface. In other embodiments, the VLP displays S protein and M protein derived antigens. In other embodiments, the VLP displays two or more domains of S protein (i.e., HR1 and HR2 domains and the linker region between the HR1 and HR2 domains) and M protein derived antigens.

(39) In order to increase the immunogenicity of the composition, in some embodiments, the immunogenic compositions comprise one or more adjuvants and/or cytokines. Suitable adjuvants include an aluminum salt such as aluminum hydroxide or aluminum phosphate, but may also be a salt of calcium, iron or zinc, or may be an insoluble suspension of acylated tyrosine, or acylated sugars, or may be cationically or anionically derivatized saccharides, polyphosphazenes, biodegradable microspheres, monophosphoryl lipid A (MPL), lipid A derivatives (e.g. of reduced toxicity), 3-O-deacylated MPL [3D-MPL], quil A, Saponin, QS21, Freund's Incomplete Adjuvant (Difco Laboratories, Detroit, Mich.), Merck Adjuvant 65 (Merck and Company, Inc., Rahway, N.J.), AS-2 (Smith-Kline Beecham, Philadelphia, Pa.), CpG oligonucleotides, bioadhesives and mucoadhesives, microparticles, liposomes, polyoxyethylene ether formulations, polyoxyethylene ester formulations, muramyl peptides or imidazoquinolone compounds (e.g. imiquamod and its homologues). Human immunomodulators suitable for use as adjuvants in the disclosure include cytokines such as interleukins (e.g. IL-1, IL-2, IL-4, IL-5, IL-6, IL-7, IL-12, etc), macrophage

colony stimulating factor (M-CSF), tumor necrosis factor (TNF), granulocyte, macrophage colony stimulating factor (GM-CSF) may also be used as adjuvants.

(40) In some embodiments, the compositions comprise an adjuvant selected from the group consisting of Montanide ISA-51 (Seppic, Inc., Fairfield, N.J., United States of America), QS-21 (Aquila Biopharmaceuticals, Inc., Lexington, Mass., United States of America), GM-CSF, cyclophosphamide, *bacillus* Calmette-Guerin (BCG), *corynebacterium parvum*, levamisole, azimezone, isoprinosine, dinitrochlorobenzene (DNCB), keyhole limpet hemocyanins (KLH), Freund's adjuvant (complete and incomplete), mineral gels, aluminum hydroxide (Alum), lysolecithin, pluronic polyols, polyanions, oil emulsions, dinitrophenol, diphtheria toxin (DT). In some embodiments, the adjuvant is Montanide adjuvant. It is expected that an adjuvant or cytokine can be added in an amount of about 0.01 mg to about 10 mg per dose, preferably in an amount of about 0.2 mg to about 5 mg per dose. Alternatively, the adjuvant or cytokine may be at a concentration of about 0.01 to 50%, preferably at a concentration of about 2% to 30%.

(41) In certain aspects, the immunogenic compositions of the disclosure are prepared by physically mixing the adjuvant and/or cytokine with peptides described herein under appropriate sterile conditions in accordance with known techniques to produce the final product. The dose may be determined according to various parameters, especially according to the substance used; the age, weight and condition of the individual to be treated; the route of administration; and the required regimen. The amount of antigen in each dose is selected as an amount which induces an immune response. A physician will be able to determine the required route of administration and dosage for any particular individual. The dose may be provided as a single dose or may be provided as multiple doses, for example taken at regular intervals, for example 2, 3 or 4 doses administered weekly. Typically, peptides, polynucleotides or oligonucleotides are typically administered in the range of 1 pg to 1 mg, more typically 1 pg to 10 µg for particle mediated delivery and 1 µg to 1 mg, and more typically 1-150 µg. Generally, it is expected that each dose will comprise 0.01-3 mg of antigen. An optimal amount for a particular vaccine can be ascertained by studies involving observation of immune responses in individuals.

(42) In preferred embodiments, the immunogenic composition contains a SARS-CoV-2 antigen VLP. Alternatively, in other embodiments, HR1 and HR2 domains of S protein and/or M protein derived antigens may exist as nucleic acids and may be formulated as a DNA vaccine. In some embodiments, DNA vaccines, or gene vaccines, comprise a plasmid with a promoter and appropriate transcription and translation control elements and a nucleic acid sequence encoding one or more polypeptides of the disclosure. In some embodiments, the plasmids also include sequences to enhance, for example, expression levels, intracellular targeting, or proteasomal processing. In some embodiments, DNA vaccines comprise a viral vector containing a nucleic acid sequence encoding one or more polypeptides of the disclosure. In additional aspects, the compositions disclosed herein comprise one or more nucleic acids encoding peptides determined to have immunoreactivity with a biological sample. For example, in some embodiments, the compositions comprise one or more nucleotide sequences encoding 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, or more peptides comprising a polypeptide or a fragment that is a fusion antigen derived from S protein and M protein of SARS-CoV-2 coronavirus. The DNA or gene vaccine also encodes immunomodulatory molecules to manipulate the resulting immune responses, such as enhancing the potency of the vaccine, stimulating the immune system or reducing immunosuppression. Strategies for enhancing the immunogenicity of DNA or gene vaccines include encoding of xenogeneic versions of antigens, fusion of antigens to molecules that activate T cells or trigger associative recognition, priming with DNA vectors followed by boosting with viral vector, and utilization of immunomodulatory molecules. In some embodiments, the DNA vaccine is introduced by a needle, a gene gun, an aerosol injector, with patches, via microneedles, by abrasion, among other forms. In some forms the DNA vaccine is incorporated into liposomes or other forms of nanobodies. In some embodiments, the DNA vaccine includes a delivery system

selected from the group consisting of a transfection agent; protamine; a protamine liposome; a polysaccharide particle; a cationic nanoemulsion; a cationic polymer; a cationic polymer liposome; a cationic nanoparticle; a cationic lipid and cholesterol nanoparticle; a cationic lipid, cholesterol, and PEG nanoparticle; a dendrimer nanoparticle. In some embodiments, the DNA vaccine is administered by inhalation or ingestion. In some embodiments, the DNA vaccine is introduced into the blood, the thymus, the pancreas, the skin, the muscle, a tumor, or other sites.

(43) In yet other embodiments, the composition may be prepared as an RNA vaccine. In some embodiments, the RNA is non-replicating mRNA or virally derived, self-amplifying RNA. In some embodiments, the non-replicating mRNA encodes the peptides disclosed herein and contains 5' and 3' untranslated regions (UTRs). In some embodiments, the virally derived, self-amplifying RNA encodes not only the peptides disclosed herein but also the viral replication machinery that enables intracellular RNA amplification and abundant protein expression. In some embodiments, the RNA is directly introduced into the individual. In some embodiments, the RNA is chemically synthesized or transcribed in vitro. In some embodiments, the mRNA is produced from a linear DNA template using a T7, a T3, or a Sp6 phage RNA polymerase, and the resulting product contains an open reading frame that encodes the peptides disclosed herein, flanking UTRs, a 5' cap, and a poly(A) tail. In some embodiments, various versions of 5' caps are added during or after the transcription reaction using a vaccinia virus capping enzyme or by incorporating synthetic cap or anti-reverse cap analogues. In some embodiments, an optimal length of the poly(A) tail is added to mRNA either directly from the encoding DNA template or by using poly(A) polymerase. In some embodiments, the fragments are derived from an antigen that is expressed within the sequence of S and/or M proteins of SARS-CoV-2. In some embodiments, the RNA includes signals to enhance stability and translation. In some embodiments, the RNA also includes unnatural nucleotides to increase the half-life or modified nucleosides to change the immunostimulatory profile.

(44) In some embodiments, the RNA is introduced by a needle, a gene gun, an aerosol injector, with patches, via microneedles, by abrasion, among other forms. In some forms the RNA vaccine is incorporated into liposomes or other forms of nanobodies that facilitate cellular uptake of RNA and protect it from degradation. In some embodiments, the RNA vaccine includes a delivery system selected from the group consisting of a transfection agent; protamine; a protamine liposome; a polysaccharide particle; a cationic nanoemulsion; a cationic polymer; a cationic polymer liposome; a cationic nanoparticle; a cationic lipid and cholesterol nanoparticle; a cationic lipid, cholesterol, and PEG nanoparticle; a dendrimer nanoparticle; and/or naked mRNA; naked mRNA with in vivo electroporation; protamine-complexed mRNA; mRNA associated with a positively charged oil-in-water cationic nanoemulsion; mRNA associated with a chemically modified dendrimer and complexed with polyethylene glycol (PEG)-lipid; protamine-complexed mRNA in a PEG-lipid nanoparticle; mRNA associated with a cationic polymer such as polyethylenimine (PEI); mRNA associated with a cationic polymer such as PEI and a lipid component; mRNA associated with a polysaccharide (for example, chitosan) particle or gel; mRNA in a cationic lipid nanoparticle (for example, 1,2 dioleoyloxy 3 trimethylammoniumpropane (DOTAP) or dioleoylphosphatidylethanolamine (DOPE) lipids); mRNA complexed with cationic lipids and cholesterol; or mRNA complexed with cationic lipids, cholesterol and PEG-lipid. In some embodiments, the RNA vaccine is administered by inhalation or ingestion. In some embodiments, the RNA is introduced into the blood, the thymus, the pancreas, the skin, the muscle, a tumor, or other sites, and/or by an intradermal, intramuscular, subcutaneous, intranasal, intranodal, intravenous, intrasplenic, intratumoral or other delivery route.

(45) In some embodiments, the polynucleotide or oligonucleotide components are naked nucleotide sequences or be in combination with cationic lipids, polymers or targeting systems. They may be delivered by any available technique. For example, the polynucleotide or oligonucleotide may be introduced by needle injection, preferably intradermally, subcutaneously or intramuscularly. Alternatively, the polynucleotide or oligonucleotide may be delivered directly across the skin using

a delivery device such as particle-mediated gene delivery. The polynucleotide or oligonucleotide may be administered topically to the skin, or to mucosal surfaces for example by intranasal, oral, or intrarectal administration. Uptake of polynucleotide or oligonucleotide constructs may be enhanced by several known transfection techniques, for example those including the use of transfection agents. Examples of these agents include cationic agents, for example, calcium phosphate and DEAE-Dextran and lipofectants, for example, lipofectam and transfectam. The dosage of the polynucleotide or oligonucleotide to be administered can be altered.

(46) The immunogenic compositions or vaccines described herein comprise, in addition to one or more peptides, nucleic acids or vectors, a pharmaceutically acceptable excipient, carrier, diluent, buffer, stabilizer, preservative, adjuvant or other materials well known to those skilled in the art. Such materials are preferably non-toxic and preferably do not interfere with the pharmaceutical activity of the active ingredient(s). The pharmaceutical carrier or diluent may be, for example, water containing solutions. The precise nature of the carrier or other material may depend on the route of administration, e.g., oral, intravenous, cutaneous or subcutaneous, nasal, intramuscular, intradermal, and intraperitoneal routes. In some embodiments, the pharmaceutical compositions of the disclosure comprise one or more “pharmaceutically acceptable carriers”. These are typically large, slowly metabolized macromolecules such as proteins, saccharides, polylactic acids, polyglycolic acids, polymeric amino acids, amino acid copolymers, sucrose (Paoletti et. al., 2001, Vaccine, 19:2118-2126), trehalose (WO 00/56365), lactose and lipid aggregates (such as oil droplets or liposomes). Such carriers are well known to those of ordinary skill in the art. In some embodiments, the pharmaceutical compositions contain diluents, such as water, saline, glycerol, etc. Additionally, auxiliary substances, such as wetting or emulsifying agents, pH buffering substances, and the like, may be present. Sterile pyrogen-free, phosphate buffered physiologic saline is a typical carrier (Gennaro et. al., 2000, Remington: The Science and Practice of Pharmacy, 20th edition, ISBN:0683306472).

(47) In some embodiments, the immunogenic compositions of the disclosure are lyophilized or in aqueous form, i.e., solutions or suspensions. Liquid formulations of this type allow the compositions to be administered direct from their packaged form, without the need for reconstitution in an aqueous medium, and are thus ideal for injection. In some embodiments, the immunogenic compositions are presented in vials, or they may be presented in ready filled syringes. The syringes may be supplied with or without needles. A syringe will include a single dose, whereas a vial may include a single dose or multiple doses. Liquid formulations of the disclosure are also suitable for reconstituting other medicaments from a lyophilized form. Where a immunogenic composition is to be used for such extemporaneous reconstitution, the disclosure provides a kit, which may comprise two vials, or may comprise one ready-filled syringe and one vial, with the contents of the syringe being used to reconstitute the contents of the vial prior to injection.

(48) In some embodiments, the immunogenic compositions of the disclosure include an antimicrobial, particularly when packaged in a multiple dose format. Antimicrobials may be used, such as 2-phenoxyethanol or parabens (methyl, ethyl, propyl parabens). Any preservative is preferably present at low levels. Preservative may be added exogenously and/or may be a component of the bulk antigens which are mixed to form the composition (e.g., present as a preservative in pertussis antigens).

(49) In some embodiments, the immunogenic compositions of the disclosure comprise detergent e.g., Tween (polysorbate), DMSO (dimethyl sulfoxide), DMF (dimethylformamide). Detergents are generally present at low levels, e.g., <0.01%, but may also be used at higher levels, e.g., 0.01-50%. In some embodiments, the immunogenic compositions of the disclosure include sodium salts (e.g., sodium chloride) and free phosphate ions in solution (e.g., by the use of a phosphate buffer).

(50) In some embodiments, the immunogenic compositions are encapsulated in a suitable vehicle either to deliver the peptides into antigen presenting cells or to increase the stability. As will be

appreciated by a skilled artisan, a variety of vehicles are suitable for delivering an immunogenic composition of the disclosure. Non-limiting examples of suitable structured fluid delivery systems may include nanoparticles, liposomes, microemulsions, micelles, dendrimers and other phospholipid-containing systems. Methods of incorporating immunogenic compositions into delivery vehicles are known in the art.

(51) Other aspects of the present invention relate to methods of inducing an immune response to at least one coronavirus polypeptide or fragment antigen as described above in a subject in need thereof, comprising the steps of: administering to the subject an effective amount of the immunogenic composition of antigenic polypeptide or fragments derived from coronavirus S protein, coronavirus M protein or combinations thereof; allowing a suitable period of time to elapse; and optionally administering at least one additional dose of the immunogenic composition. A “suitable period of time” is defined herein as a sufficient time for a subject to produce antibodies against the administered antigens described herein. A sufficient time for a subject to acquire ability to produce antibodies may be days (e.g., 2, 3, 4, 5, 6 or 7 days), weeks (e.g., 1, 2, 3 or 4 weeks), months (e.g., 1, 2, 3, 4, 5, or 6 months), or years (e.g. 1, 2, 3, 4, or 5 years) after a first, second or third dose of the immunogenic composition is administered.

(52) As used herein, the term “effective amount” refers to an amount of VLPs necessary or sufficient to realize a desired biologic effect. An effective amount of the composition would be the amount that achieves a selected result, and such an amount could be determined as a matter of routine experimentation by a person skilled in the art. For example, an effective amount for preventing, treating and/or ameliorating an infection could be that amount necessary to cause activation of the immune system, resulting in the development of an antigen specific immune response upon exposure to VLPs of the invention. The term is also synonymous with “sufficient amount” or “therapeutically effective amount”.

(53) In some embodiments, the coronavirus polypeptide or fragment antigen has a sequence identity selected from the group consisting of SEQ ID NOs:1 to 166.

(54) In some embodiments, the method of treatment comprises administration to an individual of more than one peptide, polynucleic acid in a VLP or vector. These may be administered together/simultaneously and/or at different times or sequentially. The use of combinations of different peptides, optionally targeting different antigens, may be important to overcome the challenges of viral or individual heterogeneity. The use of peptides of the disclosure in combination expands the group of individuals who can experience clinical benefit from vaccination. Multiple immunogenic compositions, manufactured for use in one regimen, may define a drug product. In some cases, different peptides, polynucleic acids or vectors of a single treatment may be administered to the individual within a period of, for example, 1 year, or 6 months, or 3 months, or 60 or 50 or 40 or 30 days.

(55) Preferably, the sequence employed to form the VLP immunogenic composition exhibits between about 60-80% sequence identity to a naturally occurring coronavirus polynucleotide or polypeptide sequence or conformational epitope sequence and more preferably the sequences exhibit between about 80-100% sequence identity, e.g., at least about 85%, 90%, 95%, 96%, 97%, 98%, or 99% sequence identity to a naturally occurring polynucleotide or polypeptide sequence or a sequence as disclosed herein. In addition, the sequences described herein can be operably linked to each other in any combinations. For example, one or more sequences may be expressed from the same promoter and/or from different promoters.

(56) Routes of administration include but are not limited to ocular, intranasal, oral, subcutaneous, intradermal, and intramuscular. The subcutaneous administration is particularly preferred. Subcutaneous administration may for example be by injection into the abdomen, lateral and anterior aspects of upper arm or thigh, scapular area of back, or upper ventrodorsal gluteal area. In some embodiments, the compositions of the disclosure are administered in one, or more doses, as well as, by other routes of administration. For example, such other routes include, intracutaneously,

intravenously, intravascularly, intraarterially, intraperitoneally, intrathecally, intratracheally, intracardially, intralobally, intramedullary, intrapulmonarily, and intravaginally. Depending on the desired duration of the treatment, the compositions according to the disclosure may be administered once or several times, also intermittently, for instance on a monthly basis for several months or years and in different dosages. Solid dosage forms for oral administration include capsules, tablets, caplets, pills, powders, pellets, and granules. In such solid dosage forms, the active ingredient is ordinarily combined with one or more pharmaceutically acceptable excipients, examples of which are detailed above. Oral preparations may also be administered as aqueous suspensions, elixirs, or syrups. For these, the active ingredient may be combined with various sweetening or flavoring agents, coloring agents, and, if so desired, emulsifying and/or suspending agents, as well as diluents such as water, ethanol, glycerin, and combinations thereof.

(57) In some embodiments, the compositions of the disclosure are administered, or the methods and uses for treatment according to the disclosure are performed, alone or in combination with other pharmacological compositions or treatments, for example other immunotherapy, vaccine or anti-viral. In some embodiments, the other therapeutic compositions or treatments are administered either simultaneously or sequentially with (before or after) the composition(s) or treatment of the disclosure.

(58) The following descriptions and examples illustrate some exemplary embodiments of the disclosed invention in detail. Those of the skill in the art will recognize that there are numerous variations and modifications of this invention that are encompassed by its scope. Accordingly, the description of a certain exemplary embodiment should not be deemed to limit the scope of the present invention.

Example 1

(59) The primary objective of designing a safer vaccine against SARS-CoV-2 using a rational approach is to minimize the possibility of inducing ADE upon vaccination, to reduce or eliminate off target penetration of the virus into immune cells and to prevent endothelial damage and thereby minimize the risk of vascular dysfunction in infected patients. A secondary objective is to create a stable formulation and affordable manufacture of the vaccine by using appropriate adjuvants and vaccine stabilization techniques.

(60) The receptor binding domain (RBD) in the spike protein is the most variable part of the virus genome (Anderson et. al., 2020). Blue spheres in FIG. 2 depict the variability of the RBD as well as the N-terminal domain (NTD) of S protein monomer based on sequence analysis of approximately 60 clinical isolates (Wrapp et. al., 2020). This RBD variability is thought to allow SARS-CoV-2 gain antigen drift, thereby permitting immune escape, while still allowing high-affinity binding to ACE2 receptor (Yang et. al., 2005). Surprisingly, convalescent plasma of COVID-19 patients who recovered without hospitalization do not contain high levels of neutralizing antibodies against RBD (Robbiani et. al., 2020). As discussed above, should variability in regions of S protein produce non-neutralizing or sub-neutralizing antibodies, that might prove detrimental through induction of ADE.

(61) In contrast, the C-terminal region of the S protein is relatively conserved, not only among distinct SARS-CoV-2 strains isolated till date, but also across other members of the sarbecovirus family members as shown below (FIG. 3, Wec et. al., 2020). The SARS-CoV-2 HR1 domain (aa 908-986) has 7 amino acids different with SARS-CoV-1 or bat coronavirus RaTG13, while HR2 (aa 1107-1175) is 100% identical at amino acid level between the three. The CH domain (aa 986-1035) is also 100% conserved whereas the CD domain (aa 1076-1141) has 9 amino acid changes against the SARS-CoV-1 or bat coronavirus RaTG1 sequences (Ceraolo et. al., 2020).

(62) The S protein is activated by proteolytic cleavage at two sites, S1/S2 and S2' (FIG. 1) by both TMPRSS2 as well as the proprotein convertase, furin (Bestle et. al., 2020; Shang et. al., 2020). The S1 subunit contains the RBD while the S2 subunit contains the hydrophobic heptad repeats HR1 and HR2 that are essential for membrane fusion. Once the RBD binds to the ACE2 receptor on

target cells, the HR1 and HR2 domains interact with one another forming a six-helix bundle complex. The connector domain between HR1 and HR2 forms a loop inducing a conformational change that allows the fusion peptide on the virus insert into the target cell membrane causing virus-cell fusion.

(63) Membrane protein (M) is a type III transmembrane glycoprotein and is the most abundant protein in SARS-CoV-2. The M protein has a short amino terminus domain located outside the virus, three transmembrane domains and a carboxy terminus domain located inside the viral envelope. On the viral surface, M protein is juxtaposed alongside the S protein and plays a role in the budding process (Alsaadi et. al., 2019). In alpha coronavirus, the M protein has been shown to participate in viral entry into host cells by facilitating membrane fusion (Naskalska et. al., 2019). While the M protein is highly conserved within SARS-CoV-2 family, there is sequence heterogeneity at the N-terminus of the protein where an insertion of a serine residue is unique to SARS-CoV-2 compared to bat or pangolin homologs (Bianchi et. al., 2020). Such mutations at the amino terminus region of the M protein could probably play a role in the host cell interactions.

(64) One aim of the present disclosure is to prevent viral fusion with cell membrane by targeting a vaccine immune response against conserved C-terminal membrane-proximal fragments of SARS-CoV-2, specifically the HR1 and HR2 regions and the spacer in between.

(65) Such a targeted approach has several advantages. First, by limiting the exposure of the host immune system, unwanted antibodies against undesirable regions of the S protein are eliminated. This in turn proves advantageous since virus bound, non-neutralizing antibodies cannot be taken up in immune effector cells such as macrophages, dendritic cells and T-cells, thereby diminishing the possibility of unwanted cytokine production and immunopathology. Second, by presenting short, rationally designed antigens to block viral fusion, the off-target entry of virus into immune cells or vascular endothelial cells can be prevented. Third, simple antigens lend themselves to simpler, cost-effective production with the possibility of easier scale up but without the need of BSL3 manufacturing and testing facilities, thereby lowering cost of goods. Fourth, antigen design will allow facile use of established adjuvants such as alum to skew host response away from a Th2 type which is linked to eosinophil-derived immunopathology (Chen et. al., 2020). The simpler, rationally designed antigens will be possessing a much better shelf life, enabling easy storage and distribution of vaccine with the need of complex logistics that are poorly available in low-income countries.

(66) Human antibodies isolated from transgenic mice against the SARS-CoV-1 S2 domain have been shown to successfully neutralize pseudo-typed viruses expressing different S proteins of various clinical isolates on SARS-CoV-1 in an RBD-independent manner (Elshabrawy et. al., 2012). In addition, monoclonal antibodies raised against SARS-CoV-1 S protein were found to neutralize viral infection by inhibiting virus entry into Vero E6 cells, a mechanism distinct from virus-ACE2 receptor binding blockade, demonstrating the feasibility of preventing viral fusion as a viable vaccine strategy (Lai et. al., 2005, Lip et. al., 2006). Similarly, anti-M protein monoclonal antibodies have been identified from lymphocytes of convalescent patients infected with SARS-CoV-1 that potently neutralize viral entry into Vero-E6 cells (Liang et. al., 2005). In addition, sera from SARS-CoV-1 have been shown to bind both amino as well as carboxy terminus of M protein (Hu et. al., 2003; Vob et. al., 2009).

Example 2

(67) Antigen Design

(68) One of the important aspects of the antigen design is to represent the prefusion state for the S protein. This is because conversion of the S protein to post-fusion state involves a significant conformational change wherein the HR1 and HR2 domains pack against one another to form an anti-parallel six-helix bundle. This rearrangement triggers fusion of the viral membrane with the host cell. Thus, the rational design strategy is to block the virus in the prefusion state and prevent conversion into the post fusion state. In such a case, the viral entry and spread can be stopped

irrespective of the cell the virus infects. Indeed, it has been shown that the SARS-CoV-1 HR2 domain forms a coiled coil structure in solution consisting of three helices folded as a parallel trimer, as in prefusion state (Hakkanson et. al., 2006). Thus, even targeting HR2 as a standalone antigen may be possible.

(69) Seven distinct antigens were designed based on M protein and the heptad repeat structure within the prefusion SARS-CoV-2 S protein as follows: SC2MN1: Amino acids 1-31 covering amino terminus of M protein SC2MC1: Amino acids 130-160 from carboxy terminus of M protein SC2SHR1: Amino acids 913-984 covering the first heptad repeat HR1 of S protein SC2HSHRL: Amino acids 1147-1170 covering the linker region connecting HR1 and HR2 SC2HSHR2: Amino acids 1171-1212 covering the second heptad repeat HR2 of S protein SC2HSHR2L: Amino acids 1147-1212 covering both the linker as well as HR2 regions SC2RBD (control antigen): Amino acids 318-541 representing RBD of S protein

(70) In reference to FIG. 2, the composition of the proposed antigens in those cases where high-resolution structures of S or M protein are available (6VXX.PDB, 6LXT.PDB, 6M3W.PDB). These antigens were created in the form of fusion protein expressed on virus like particles (VLP) using the Hepatitis B surface antigen S protein HBV-S. Presenting viral antigens on VLPs will result in enhanced quality of host immune response because VLPs can present antigen in draining lymph nodes and efficiently engage B cell receptors due to their repetitive structure resulting in a higher abundance of neutralizing antibodies (Coleman et. al., 2014). The HBV-S core protein is a very well-established system for VLP generation (Martini et. al., 2019).

(71) A widely used approach for presenting antigens on VLPs is through the use of genetic fusion. However, while this technique can result in virus-like configuration of the antigen, longer peptides or those with charge or significant hydrophobicity can interfere with the actual assembly of the VLPs. To circumvent this problem, a mosaic approach was undertaken to construct VLPs (Ramasamy et. al., 2018). Specifically, the HBV S protein was coexpressed with proposed antigens in a fixed stoichiometry. In some embodiments, the ratio of HBV S protein to a SARS-CoV-2 polypeptide may be 6:1, preferably 5:1, more preferably 4:1. Microscopic as well as immunological assessment were carried out to ensure bona fide VLP formation. Because the HR1, linker and HR2 regions are conserved between SARS CoV-1 and SAR-CoV-2, monoclonal antibodies previously made using SARS-CoV-1 were used as analytical and immunological tools to ensure proper VLP assembly. Each of the antigens will be produced and tested, in vitro and in vivo.

Example 3

(72) Production of Immunogens

(73) VLPs contained within formulations of approved vaccines such as those against human papillomavirus (HPV) or hepatitis B virus (HBV) are produced in yeast cells. These vaccines have proven to be safe and effective, and their manufacturing processes are very well established. Separately, VLPs expressed in recombinant baculovirus systems covering multi-component antigens such as HA and matrix 1 for influenza vaccine have also been established. The antigens proposed in our strategy are glycosylated at multiple sites and it becomes important to ensure glycosylation in antigen preparation as close to the human system as possible (Wu et. al., 2010). Therefore, immunogen production in Vero cells is proposed (Ammerman et. al., 2008).

(74) Vero cells are derived from African green monkeys and are a close approximation to human cells. Vero cell lines have been approved in the US for production of licensed viral vaccines such as those against rotavirus, smallpox and inactivated poliovirus. Worldwide, Vero cells have also been used for the production of vaccines against Rabies virus, Reovirus and Japanese encephalitis virus. Commercially, Vero cells have been scaled up to 660 m² with cell density approaching 2.3×10⁵ cells/cm³. As such Vero cell technology, is cheap, scalable and well-established vaccine production technology worldwide.

Example 4

(75) Epitope Prediction for SARS-CoV-2 M and S Antigens

(76) 1: M protein (UniProt: P59596)

(77) The results presented below for T cell epitopes (Table 1) are derived from experimental data deposited at the IEDB ImmunoBrowser (Vita et al., 2014). Due to the unavailability of a structure for the M protein, a sequence-based algorithm for B cell epitope prediction was used (BepiPred 2.0, Table 2; Jespersen et al., 2017).

(78) TABLE-US-00019 TABLE 1 T cell epitope prediction for the N (top) and C terminus (bottom) of SARS-CoV-2 M protein based on experimental data available at the IEDB. Epitopes tested immediately downstream or upstream of these sequences did not give positive responses. SEQ Mapped Mapped ID Start End Subjects Subjects Assay Assay Response NO Sequence Position Position Tested Responded Positive Negative Freq. 1 MADNGTIT 1 15 4 1 1 1 0.25 VEELKQL 2 MADNGTIT 1 16 40 10 1 0 0.25 VEELKQLL 3 MADNGTIT 1 18 1 1 1 0 1 VEELKQL LEQ 4 MADNGTIT 1 20 32 1 1 1 0.03 VEELKQLL EQWN 5 MADNGTI 1 31 40 40 1 0 1 TVEELK QLLEQW NLVIGF LFLAWI 6 ITVEEL 7 14 4 3 1 0 0.75 KQ 7 ITVEELKQ 7 23 40 28 1 0 0.7 LLEQWNLV I 8 KLNTDHA 204 221 1 1 1 0 1 GSNDNIAL LVQ 9 TDHAGSND 207 221 48 27 1 1 0.56 NIALLVQ

Sequence-Based B Cell Epitope Prediction Using the BepiPred 2.0 Server

(79) TABLE-US-00020 TABLE 2 B cell epitope prediction for the N (left) and C terminus (center, right) of SARS-CoV-2M protein based on the BepiPred algorithm, a server for sequence-based epitope prediction (cbs.dtu.dk/services/BepiPred/). Residues immediately downstream or upstream of these were not identified as predicted epitopes (default threshold of 0.5). Position AminoAcid Exposed/Buried EpitopeProbability 5 G E 0.501 6 T E 0.522 7 I B 0.535 8 T E 0.524 9 V B 0.531 10 E E 0.539 11 E E 0.538 12 L B 0.518 13 K B 0.519 14 Q E 0.541 15 L B 0.541 16 L B 0.530 17 E E 0.540 18 Q E 0.521 180 L B 0.519 181 G E 0.530 182 A E 0.549 183 S E 0.572 184 Q E 0.587 185 R E 0.595 186 V B 0.600 187 G E 0.598 188 T E 0.576 189 D E 0.555 190 S B 0.540 191 G B 0.522 192 F B 0.501 195 Y B 0.505 196 N B 0.509 197 R B 0.531 198 Y E 0.549 199 R E 0.560 200 I B 0.571 201 G E 0.567 202 N E 0.564 203 Y B 0.563 204 K E 0.563 205 L B 0.580 206 N E 0.594 207 T B 0.606 208 D E 0.613 209 H E 0.621 210 A E 0.630 211 G E 0.636 212 S E 0.626 213 N E 0.617 214 D E 0.594 215 N E 0.568 216 I B 0.538 217 A E 0.506

2: S Protein (UniProt: P59594)

The results presented below for T cell epitopes (Tables 3 and 5) are derived from experimental data available at the IEDB ImmunoBrowser (Vita et al., 2014). Structure-based B cell epitope predictions (Tables 4 and 6) were generated with SEPPA 3.0 (Zhou et al., 2019). Table 7 shows a sequence-based epitope prediction using BepiPred 2.0 (Jespersen et al., 2017) for amino acids 1104-1184 as the structures available did not span that region.

(80) TABLE-US-00021 TABLE 3 T cell epitope prediction for the receptor binding domain of SARS-CoV-1 S protein (residues 306-527) with peptides spanning the receptor-binding motif (424-494) highlighted in blue. Response frequency is based on experimental data available at the IEDB. Tested epitopes that did not give positive responses are absent. SEQ Mapped Mapped ID Start End Subjects Subjects Assay Assay Response NO Sequence Position Position Tested Responded Positive Negative Freq. 10 EIDKGIYQTS 294 310 42 21 1 0 0.5 NFRVVPS 11 KGIYQTSNFR 297 316 1 1 1 0 1 VVPSGDVVR F 12 KGIYQTSN 297 304 4 2 1 0 0.5 13 IYQTSNFRVV 299 316 3 1 1 2 0.33 PSGDVVRF 14 TSNFRVVPS 302 318 42 20 1 0 0.48 GDVVRFPN 15 SGDVVRFPNI 310 326 42 13 TNLCPPFG 1 0 0.31 16 VRFPNITNLC 314 330 42 10 1 0 0.24 PFGEVFN 17 LCPFGGEVFN 322 338 42 11 1 0 0.26 ATKFPSVY 18 FGEVFNAT 325 332 4 2 1 0 0.5 19 FNATKFPSV 329 345 3 2 YAWERKKI 1 0 0.67 N330, A331, 330 485 1 1 1 0 1 T332, K333, F360, W423, N424, N427, I428, A430, T431, S432, N435, N437, T485 20 TKFPSVYAW 332 351 1 1 1 0 1 ERKKISNCV AD 21 PSVYAWERK 335 351 42 12 1 0 0.29 KISNCVAD 22 SVYAWERK 336 352 3 1 KISNCVADY 1 2 0.33 23 VYAWERKKI SNCVADYSV 337 360 1 1 1 0 1 LYNSTF K343, K344, 343 428 1 1 1 0 1 1345, S346,

N347, C348, V349, A350, D351, Y352, S353, V354, L355, N356, N357, S358,
T359, F360, F361, S362, T363, F364, K365, C366, Y367, K373, L374, N375, D376,
L377, C378, F379, S380, N381, V382, Y383, A384, D385, S386, F387, V388, V389,
K390, K411, L412, P413, D414, D415, F416, M417, G418, C419, V420, L421,
A422, W423, N424, T425, R426, N427, I428 24 KKISNCVAD 343 360 5 1 1 2 0.2
YSVLNSTF 25 KKISNCVAD 343 359 42 5 1 0 0.12 YSVLYNST K344, F360, 344 487 1 1 1 0
1 Y442, L472, D480, T487 26 CVADYSVLY 348 356 11 3 2 0 0.27 27 YSVLYNSTFF 352
368 42 12 1 0 0.29 STFKCYG Y356, N357, 356 494 1 1 1 0 1 S358, T359, F361, S362,
T363, F364, K365, C366, A371, T372, G391, R395, N424, 1489, Y494 Y356, N357,
356 417 1 1 1 0 1 S362, T363, F364, K365, C366, Y367, G368, V369, S370, A371,
T372, K373, D376, L377, R395, D415, M417 28 STFFSTFKCY 358 374 4 2 2 2 0.5
GVSATKL T359, S362, 359 494 1 1 1 0 1 G391, D392, N424, R426, N427, T486, T487,
G488, 1489, G490, Y491, Q492, Y494 T359, T363, 359 494 2 1 1 1 0.5 K365, K390,
G391, D392, R395, R426, Y436, G482, Y484, T485, T486, T487, G488, 1489, G490,
Y491, Q492, Y494 29 FFSTFKCYG 360 376 42 5 1 0 0.12 VSATKLND 30 KCYGVSATK 365
374 4 3 3 1 0.75 L 31 CYGVSATKL 366 374 3 3 3 0 1 32 CYGVSATKL 366 382 42 13 1 0 0.31
NDLCFSNV 33 YGVSATKL 367 374 1 1 1 0 1 34 KLNDLCFSN 373 389 42 8 1 0 0.19
VYADSFVV 35 SNVYADSFV 380 399 1 1 1 0 1 VKGDDVRQI AP 36 NVYADSFVV 381 397
43 13 KGDDVRQI 1 1 0.3 D392, V394 392 394 1 1 1 0 1 D392, V394, 392 454 1 1 1 0 1
D414, F416, R441, D454 37 IAPGQTGVIA 397 413 42 10 DYNKLP 1 0 0.24 38
IADYNKLP 405 421 42 5 DDFMGCVL 1 0 0.12 39 KLPDDFMGC 411 427 42 9 VLAWNTRN
1 0 0.21 40 KLPDDFMGC 411 420 20 4 V 1 1 0.2 41 CVLAWNTR 419 435 42 8 1 0 0.19
NIDATSTGN 42 NTRNIDATST 424 435 8 8 1 0 1 GN R426, S432, 426 492 1 1 1 0 1 T433,
Y436, N437, K439, Y440, Y442, P469, P470, A471, L472, N473, C474, Y475, W476,
L478, N479, D480, Y481, G482, Y484, T485, T486, T487, G488, 1489, Y491, Q492 43
NIDATSTGN 427 444 3 1 1 2 0.33 YNYKYRYLR 44 NIDATSTGN 427 443 42 11 1 0 0.26
YNYKYRYL I428, A430, 428 439 1 1 K439 1 0 1 D429, R441, 429 454 1 1 D454 1 0 1
D429, R441, 429 454 1 1 E452, D454 1 0 1 D429, R441, 429 454 1 1 D454 1 0 1 D429,
R441, 429 463 1 1 D454, D463 1 0 1 D429, R441, 429 454 1 E452, D454 1 1 0 1 T431,
S432, 431 446 1 1 K439, G446 1 0 1 45 TSTGNYNKY 431 445 4 1 1 1 0.25 YRYLRH 46
GNYNYKYR 434 448 1 1 1 0 1 YLRHGKL 47 GNYNYKYR 434 453 1 1 1 0 1 YLRHGKLRP
FER 48 NNYNYKYRYL 435 444 3 2 2 1 0.67 R 49 NNYNYKYRYL 435 451 46 12 3 2 0.26
RHGKL RPF 50 YNYKYRYL 436 443 2 2 2 0 1 51 YNYKYRYLR 436 455 1 1 1 0 1
HGKL RPFER DI 52 HNYKYRYL 436 443 18 12 3 0 0.67 53 YNYKYRYLR 436 450 4 1
HGKL RPF 1 1 0.25 54 NYKYRYLRH 437 459 1 1 GKLRPFER DI SNVP 1 0 1 55 KYRYLRHGK
439 454 1 1 LRPFER DI 1 0 1 K439, G446, 439 463 1 1 S461, D463 1 0 1 R441 441 441 2 2 2 0
1 Y442 442 442 1 1 1 0 1 Y442, A834 442 834 1 1 1 0 1 Y442, D757 442 757 1 1 1 0 1 56
YLRHGKL RPF 442 458 4 2 2 2 0.5 FERDISNV 57 YLRHGKL RPF 442 459 4 2 1 0 0.5
FERDISNV 58 LRHGKL RPF 443 459 3 2 1 0 0.67 ERDISNV G446, P462, 446 475 1 1 1 0 1
D463, Y475 59 KLRPFER DI 447 455 1 1 1 0 1 60 KLRPFER DI 447 458 1 1 1 0 1 NV 61
RPFERDISNV 449 458 2 2 2 0 1 62 RPFERDISNV 449 461 4 2 1 0 0.5 PFS 63 RPFERDISNV
449 465 44 11 1 2 0.25 PFSPDGK D454 454 454 1 1 1 0 1 64 SNVPFSPDG 456 472 44 11 1 2
0.25 KPCTPPAL 65 PFSPDGKPCT 459 470 8 8 1 0 1 PP 66 FSPDGKPCTP 460 476 1 1 1 0 1
PALNCYW 67 SPDGKPCTPP 461 477 42 5 1 0 0.12 ALNCYWP P462 462 462 1 1 1 0 1 68
CTPPALNCY 467 483 42 8 1 0 0.19 WPLNDYGF 69 ALNCYWPLN 471 503 1 1 1 0 1
DYGFYTTTGI GYQPYRVVV LSFEL 70 ALNCYWPLN 471 503 42 30 1 0 0.71 DYGFYTTTGI
GYQPYRVVV LSFEL 71 ALNCYW 471 476 4 1 1 0 0.25 72 CYWPLNDY 474 490 42 7 1 0 0.17
GFYTTTGI N479 479 479 1 1 1 0 1 D480, Y484 480 484 1 1 1 0 1 73 GFYTTTGI GY 482 498
42 8 1 0 0.19 QPYRVVV Y484, T487 484 487 1 1 1 0 1 74 TTGIGYQ 486 492 1 1 1 0 1 75
GYQPYRVVV 490 510 2 2 2 0 1 LSFELLNAPA TV 76 GYQPYRVVV 490 506 44 9 2 1 0.2

LSFELLNA 77 QPYRVVLS 492 501 4 1 1 0 0.25 F 78 VVLSFELLN 497 514 2 1 1 1 0.5
APATVCGPK 79 VLSFELLNAP 498 514 42 8 1 0 0.19 ATVCGPK 80 NAPATVCGP 505 521 44
7 1 2 0.16 KLSTDLIK 81 CGPKLSTDLI 511 527 42 7 1 0 0.17 KNQCVNF

Sequence-Based B Cell Epitope Prediction Using the SEPPA 3.0 Server and the Atomic
Coordinates Deposited as 5X58.Pdb.

(81) TABLE-US-00022 TABLE 4 B cell epitope prediction for the receptor binding domain of
SARS-CoV-2 S protein (residues 306-527) with peptides spanning the receptor-binding motif (424-
494) highlighted in blue. The server SEPPA 3.0 was used with the default threshold of 0.064.

Absent residues did not rank as epitope residues. resS eq resName score location 314 VAL 0.100
surface 315 ARG 0.081 surface 316 PHE 0.152 surface 317 PRO 0.101 surface 318 ASN 0.155
surface 331 ALA 0.115 surface 332 THR 0.158 surface 334 PHE 0.097 surface 335 PRO 0.110
surface 336 SER 0.083 surface 337 VAL 0.073 surface 338 TYR 0.077 surface 347 ASN 0.127
surface 378 CYS 0.179 surface 379 PHE 0.104 surface 380 SER 0.258 surface 381 ASN 0.072
surface 382 VAL 0.070 surface 391 GLY 0.073 surface 400 GLY 0.071 surface 401 GLN 0.068
surface 403 GLY 0.071 surface 405 ILE 0.085 surface 406 ALA 0.073 surface 407 ASP 0.079
surface 408 TYR 0.097 surface 409 ASN 0.080 surface 411 LYS 0.083 surface 424 ASN 0.066
surface 425 THR 0.072 surface 426 ARG 0.162 surface 427 ASN 0.078 surface 428 ILE 0.084
surface 429 ASP 0.083 surface 430 ALA 0.122 surface 431 THR 0.174 surface 432 SER 0.222
surface 433 THR 0.145 surface 434 GLY 0.202 surface 435 ASN 0.166 surface 436 TYR 0.220
surface 437 ASN 0.096 surface 438 TYR 0.101 surface 439 LYS 0.181 surface 440 TYR 0.081
surface 441 ARG 0.071 surface 442 TYR 0.093 surface 443 LEU 0.105 surface 444 ARG 0.087
surface 445 HIS 0.118 surface 446 GLY 0.073 surface 447 LYS 0.075 surface 448 LEU 0.066
surface 451 PHE 0.067 surface 454 ASP 0.083 surface 456 SER 0.114 surface 457 ASN 0.117
surface 458 VAL 0.116 surface 459 PRO 0.110 surface 460 PHE 0.130 surface 461 SER 0.171
surface 462 PRO 0.165 surface 463 ASP 0.173 surface 464 GLY 0.185 surface 465 LYS 0.174
surface 466 PRO 0.178 surface 467 CYS 0.161 surface 468 THR 0.195 surface 469 PRO 0.189
surface 470 PRO 0.225 surface 471 ALA 0.233 surface 472 LEU 0.242 surface 473 ASN 0.186
surface 474 CYS 0.193 surface 475 TYR 0.116 surface 476 TRP 0.106 surface 478 LEU 0.179
surface 479 ASN 0.172 surface 480 ASP 0.100 surface 481 TYR 0.144 surface 482 GLY 0.255
surface 483 PHE 0.138 surface 484 TYR 0.236 surface 485 THR 0.206 surface 486 THR 0.203
surface 487 THR 0.238 surface 488 GLY 0.210 surface 489 ILE 0.115 surface 490 GLY 0.105
surface 491 TYR 0.117 surface 492 GLN 0.096 surface 494 TYR 0.070 surface 501 PHE 0.072
surface 502 GLU 0.254 surface 503 LEU 0.261 surface 504 LEU 0.434 surface 505 ASN 0.324
surface 506 ALA 0.362 surface 507 PRO 0.441 surface 508 ALA 0.259 surface 509 THR 0.147
surface

Heptad Repeat 1 (902-952), Heptad Repeat 2 (1145-1184) and a Linker T Cell Epitopes Based on
Data Available at the IEDB immunoBrowser.

(82) TABLE-US-00023 TABLE 5 T cell epitope prediction for heptad repeat 1,
linker and heptad repeat 2 of SARS- CoV-2 S protein (residues 902-1184).

Response frequency is based on experimental data available at the IEDB.

Tested epitopes that did not give positive responses are absent. SEQ Mapped
Mapped ID Start End Subjects Subjects Assay Assay Response NO Sequence Position Position
Tested Responded Positive Negative Freq. 82 VLYENQK 897 913 43 20 1 1 0.47 QIANQFNK AI
83 QIANQFNK 904 920 43 5 1 1 0.12 AISQIQESL 84 KAISQIQES 911 927 43 6 1 1 0.14
LTTTSTAL Q917, E918, 917 940 1 1 1 0 1 S919, T921, T922, S924, T925, A926, G928,
K929, Q931, D932, V933, N935, Q936, A938, Q939, A940 85 ESLTTTSTA 918 934 45 13
2 1 0.29 LGKLQDV V T921, T922, 921 944 1 1 1 0 1 T923, T925, A926, G928, K929,
L930, D932, V933, N935, Q936, N937, Q939, A940, N942, T943, L944 86 TALGKLQD
925 941 43 8 1 1 0.19 VVNQNAQ AL 87 DVVNQNA 932 948 43 9 1 1 0.21 QALNTLVK QL 88
QALNTLVK 939 955 46 17 2 2 0.37 QLSSNFGAI 89 ALNTLVKQ 940 948 52 3 3 4 0.06 L 90

KQLSSVLNDI 946 962 3 1 1 0 0.33 AISSVLNDI 91 AISSVLNDI 954 971 1 1 1 0 1 LSRLDKVE A
 92 AISSVLNDI 954 970 42 11 1 0 0.26 LSRLDKVE 93 SVLNDILSR 957 965 13 1 1 1 0.08 94
 VLNDILSR 958 966 166 12 6 3 0.07 L 95 ILSRLDKV 962 978 3 1 1 0 0.33 EAEVQIDR L 96
 RLDKVEAE 965 973 41 5 2 2 0.12 V 97 EAEVQIDR 970 986 43 6 1 1 0.14 LITGRLQSL 98
 AEVQIDRLI 971 979 5 1 1 0 0.2 RLITGRLQS 99 LQTYVTQQ 977 993 42 19 1 0 0.45 100
 LITGRLAA 978 986 2 1 1 1 0.5 L 101 LITGRLQSL 978 986 67 5 5 5 0.07 102 RLQSLQTY 982
 990 63 18 4 3 0.29 V 103 SLQTYVTQ 985 1001 42 16 1 0 0.38 QLIRAAEIR 104 QLIRAAEIR
 993 1010 4 3 3 0 0.75 ASANLAAT K 105 QLIRAAEIR 993 1009 42 8 1 0 0.19 ASANLAAT 106
 RASANLAA 1001 1017 42 7 1 0 0.17 TKMSECVL G 107 AATKMSEC 1007 1023 42 13 1 0 0.31
 VLGQSKRV D 108 VLGQSKRV 1015 1031 3 1 1 0 0.33 DFCGKGY HL 109 RVDFCGKG 1021
 1029 36 2 1 0 0.06 Y 110 DFCGKGY 1023 1039 42 11 1 0 0.26 HLMSFPQA AP 111 LMSFPQAA
 1031 1047 44 14 2 1 0.32 PHGVVFLH V 112 APHGVVFL 1038 1047 13 3 1 0 0.23 HV 113
 PHGVVFLH 1039 1055 42 11 1 0 0.26 VTYVPSQE R 114 GVVFLHVT 1041 1049 2 1 1 0 0.5 Y
 115 VVFLHVTY 1042 1050 37 4 4 2 0.11 V 116 VTYVPSQE 1047 1063 42 10 1 0 0.24
 RNFTTAPAI 117 ERNFTTAP 1054 1071 2 1 1 1 0.5 AICHEGKA YF 118 RNFTTAPAI 1055 1071
 42 7 1 0 0.17 CHEGKAYF 119 PAICHEGK 1061 1093 15 12 1 0 0.8 AYFPREGV FVFNGTSW
 FITQRNFFS 120 HEGKAYFP 1065 1076 8 8 1 0 1 REGV 121 AYFPREGV 1069 1085 42 2 1 0
 0.05 FVFNGTSW F 122 FVFNGTSW 1077 1093 42 2 1 0 0.05 FITQRNFFS 123 GTSWFITQ
 1081 1095 3 1 1 0 0.33 RNFFSPQ 124 SWFITQRN 1083 1097 5 3 2 1 0.6 FFSPQII 125
 NFFSPQIIT 1090 1106 3 1 1 0 0.33 TDNTFVSG 126 FFSPQIITT 1091 1130 1 1 1 0 1
 DNTFVSGN 127 CDVVIGIIN 1096 1104 18 1 1 0 0.06 NTVYDPLQ PELDSF IITTDNTFV 128
 ITTDNTFVS 1097 1113 3 1 1 0 0.33 GNCDVVIG 129 FVSGNCDV 1103 1120 2 1 1 1 0.5
 VIGIINNTV Y 130 SGNCDVVI 1105 1121 42 6 1 0 0.14 GIINNTVY D 131 VIGIINNTV 1111
 1130 2 2 2 0 1 YDPLQPEL DSF 132 VIGIINNTV 1111 1127 44 9 2 1 0.2 YDPLQPEL 133
 TVYDPLQP 1118 1134 43 2 1 1 0.05 ELDSFKEE L 134 PELDSFKE 1125 1141 4 1 1 1 0.25
 ELDKYFKN H 135 DSFKEELD 1128 1159 2 1 1 1 0.5 KYFKNHTS PDVDLGD I SGINASVV
 136 DSFKEELD 1128 1137 36 1 1 0 0.03 KY 137 EELDKYFK 1132 1148 4 1 1 1 0.25
 NHTSPDVD L 138 DKYFKNHT 1135 1150 1 1 1 0 1 SPDVDLGD 139 KNHTSPDV 1139 1155
 42 4 1 0 0.1 DLGDISGIN 140 SPDVDLGD 1143 1157 65 2 2 2 0.03 ISGINAS 141 VDLGDISGI
 1146 1154 1 1 1 0 1 142 DLGDISGIN 1147 1163 43 6 1 1 0.14 ASVVNIQK 143 LGDISGINA
 1148 1162 17 1 1 0 0.06 SVVNIQ 144 ISGINASVV 1151 1170 1 1 1 0 1 NIQKEIDRL NE 145
 ISGINASVV 1151 1192 1 1 1 0 1 NIQKEIDRL NEVAKNL NESLIDLQE LGKYEQYI N1155,
 1155 1178 1 1 1 0 1 A1156, S1157, V1159, N1160, Q1162, K1163, E1164, D1166, R1167, N1169,
 E1170, V1171, K1173, N1174, N1176, E1177, S1178 Q1162, 1162 1185 1 1 1 0 1 K1163, E1164,
 D1166, R1167, N1169, E1170, V1171, K1173, N1174, N1176, E1177, S1178, 11180, D1181,
 Q1183, E1184, L1185 146 KEIDRLNE 1163 1179 3 1 1 0 0.33 VAKNLNES L 147 EIDRLNEV
 1164 1191 1 1 1 0 1 AKNLNESL IDLQELGK YEQY 148 EIDRLNEV 1164 1191 42 28 1 0 0.67
 AKNLNESL IDLQELGK YEQY 149 RLNEVAKN 1167 1175 246 17 7 6 0.07 L 150 EVAKNLNE
 1170 1186 42 6 1 0 0.14 SLIDLQELG

Structure-based B cell epitope prediction using the SEPPA 3.0 server and the atomic coordinates deposited as 5X58.pdb: For amino acids downstream of 1104, a sequence-based method (BepiPred 2.0) was used due to the lack of a solved structure encompassing that region.

(83) TABLE-US-00024 TABLE 6 B cell epitope prediction for heptad repeat 1 (residues 902-952) and linker up to residue 1104 of SARS-CoV-2 S protein (remaining linker region and heptad repeat 2 are absent from the atomic coordinates of 5X58.pdb). The server SEPPA 3.0 was used with the default threshold of 0.064. Absent residues did not rank as epitope residues. resSeq resName score location 902 GLN 0.702 surface 903 LYS 0.515 surface 904 GLN 0.486 surface 906 ALA 0.430 surface 907 ASN 0.313 surface 908 GLN 0.302 surface 909 PHE 0.146 surface 910 ASN 0.136 surface 912 ALA 0.089 surface 913 ILE 0.122 surface 914 SER 0.104 surface 915 GLN 0.098 surface 916 ILE 0.068 surface 917 GLN 0.073 surface 918 GLU 0.099 surface 919 SER 0.109

surface 920 LEU 0.084 surface 921 THR 0.104 surface 941 LEU 0.097 surface 942 ASN 0.134
 surface 943 THR 0.134 surface 944 LEU 0.107 surface 945 VAL 0.113 surface 947 GLN 0.174
 surface 948 LEU 0.222 surface 949 SER 0.151 surface 950 SER 0.157 surface 952 PHE 0.236
 surface 953 GLY 0.183 surface 954 ALA 0.172 surface 955 ILE 0.097 surface 956 SER 0.080
 surface 957 SER 0.117 surface 958 VAL 0.089 surface 959 LEU 0.258 surface 960 ASN 0.115
 surface 961 ASP 0.111 surface 962 ILE 0.115 surface 963 LEU 0.090 surface 964 SER 0.147
 surface 965 ARG 0.144 surface 966 LEU 0.133 surface 967 ASP 0.135 surface 968 LYS 0.208
 surface 969 VAL 0.177 surface 970 GLU 0.194 surface 972 GLU 0.236 surface 973 VAL 0.366
 surface 974 GLN 0.237 surface 975 ILE 0.395 surface 976 ASP 0.401 surface 977 ARG 0.283
 surface 978 LEU 0.270 surface 979 ILE 0.413 surface 980 THR 0.432 surface 981 GLY 0.360
 surface 982 ARG 0.240 surface 983 LEU 0.288 surface 984 GLN 0.344 surface 985 SER 0.176
 surface 987 GLN 0.098 surface 988 THR 0.070 surface 1028 GLY 0.149 surface 1029 TYR 0.255
 surface 1050 VAL 0.513 surface 1051 PRO 0.432 surface 1052 SER 0.479 surface 1053 GLN
 0.427 surface 1054 GLU 0.159 surface 1058 THR 0.073 surface 1059 THR 0.107 surface 1060
 ALA 0.135 surface 1061 PRO 0.131 surface 1062 ALA 0.261 surface 1063 ILE 0.245 surface 1064
 CYS 0.146 surface 1065 HIS 0.230 surface 1066 GLU 0.249 surface 1067 GLY 0.237 surface 1068
 LYS 0.211 surface 1069 ALA 0.268 surface 1070 TYR 0.218 surface 1071 PHE 0.274 surface 1072
 PRO 0.297 surface 1073 ARG 0.269 surface 1074 GLU 0.214 surface 1075 GLY 0.201 surface
 1076 VAL 0.085 surface 1079 PHE 0.091 surface 1080 ASN 0.093 surface 1082 THR 0.090
 surface 1083 SER 0.103 surface 1084 TRP 0.076 surface 1085 PHE 0.080 surface 1086 ILE 0.137
 surface 1087 THR 0.170 surface 1088 GLN 0.147 surface 1089 ARG 0.168 surface 1090 ASN
 0.253 surface 1091 PHE 0.398 surface 1092 PHE 0.136 surface 1093 SER 0.169 surface 1094 PRO
 0.105 surface 1095 GLN 0.130 surface 1096 ILE 0.146 surface 1097 ILE 0.210 surface 1098 THR
 0.233 surface 1099 THR 0.328 surface 1100 ASP 0.376 surface 1101 ASN 0.273 surface 1102 THR
 0.321 surface 1103 PHE 0.322 surface 1104 VAL 0.320 surface

(84) TABLE-US-00025 TABLE 7 B cell epitope prediction for amino acids 1105-1184 of SARS-CoV-2 S comprising the heptad repeat 2 and part of the upstream linker protein (based on the BepiPred algorithm, a server for sequence-based analysis (cbs.dtu.dk/services/BepiPred/)). Residues immediately downstream or upstream of these were not identified as predicted epitopes (default threshold of 0.5). Position AminoAcid Exposed/Buried EpitopeProbability 1115 I B 0.515 1116 N E 0.522 1117 N E 0.548 1118 T E 0.567 1119 V E 0.575 1120 Y B 0.586 1121 D E 0.589 1122 P E 0.582 1123 L B 0.569 1124 Q B 0.557 1125 P E 0.559 1126 E E 0.541 1127 L E 0.529 1128 D B 0.521 1129 S E 0.531 1130 F B 0.515 1131 K E 0.508 1132 E E 0.517 1133 E E 0.527 1134 L B 0.519 1135 D E 0.544 1136 K E 0.555 1137 Y B 0.569 1138 F B 0.576 1139 K E 0.587 1140 N E 0.595 1141 H B 0.601 1142 T E 0.604 1143 S E 0.617 1144 P E 0.601 1145 D E 0.606 1146 V B 0.607 1147 D E 0.590 1148 L B 0.600 1149 G E 0.593 1150 D E 0.572 1151 I B 0.574 1152 S E 0.564 1153 G E 0.560 1154 I B 0.535 1155 N E 0.516 1156 A B 0.517 1177 E E 0.501 1180 I B 0.508 1181 D B 0.510 1182 L B 0.511 1183 Q E 0.517 1184 E E 0.509

Example 5

(85) Biochemical and Cell Based Assays

(86) Immune sera from vaccinated animals will be tested using ELISA. Here, the binding of polyclonal sera derived from animals is tested against VLP antigen immobilized on wells using serial dilutions. Binding of ELISA positive sera will be tested against full length SARS-COV-2 S protein. This will be done in two ways. Using ELISA, binding of anti-sera will be verified against the ectodomain of S protein recombinantly-expressed in 293HEK cells and purified to homogeneity using affinity chromatography. Further, binding will be characterized against full length S protein transiently-expressed in CHO or COS-7 cells using flow cytometry. These assays address whether anti-sera from immunized animals bind prefusion, glycosylated and intact S protein. Finally, neutralizing activity of antisera will be verified using live virus neutralization assays wherein SARS-CoV-2 virus together with antisera are used to infect Vero cells followed by detection of

internalized virus using commercially available, anti-S protein antibody such as CR3022 (Pinto et. al., 2020).

Example 6

(87) Animal Studies

(88) Mouse studies will be conducted to screen the antigen panel and identify a lead and backup vaccine candidate. Up to 3 dose strengths will be tested for each vaccine construct. All vaccine formulations will be alum adjuvanted. Groups of mice (n=10, 6-8 week old female Balb/c) will be vaccinated with VLP vaccine constructs intramuscularly followed by a booster at 14 day intervals. Sera will be collected prior to immunization as well after 7 days after each immunization cycle. 10 days after the booster, animals will be challenged intranasal with SARS-CoV-2 virus. Animals will be monitored for signs of disease. Animals will be euthanized at the end of the study and lung tissue are harvested for pathology and immunohistochemical staining.

(89) Non-Human Primate (NHP) Studies:

(90) NHP studies will be carried out on the lead and backup vaccine candidates. The lead and/or the backup mixture of more than one of the vaccine constructs, including the RBD construct, will be also considered for the study. Up to 2 doses (high dose and low dose) for each vaccine construct will be tested. All vaccine formulations will be alum adjuvanted. Adult rhesus macaques (n=4 for each group, M/F) will be vaccinated be intramuscularly followed by a booster 14 days later. Blood samples will be collected before and after vaccination. Seven days following the booster, animals will be challenged with live SARS-CoV-2 virus intratracheally, intranasally, orally and ocularly. Clinical exams will be performed at regular intervals and BAL fluid are collected on periodic days. At the end of the study, animals will be euthanized and necropsy are performed.

(91) Vaccine Safety Evaluation:

(92) Safety evaluation of the lead vaccine candidate will be conducted in macaques. Groups of animals (n=5 M/F) will be immunized with high dose of the lead vaccine. A total of 3 intramuscular injections will be made (immunization plus two boosters). Safety evaluation include clinical observations as well as gross pathology on lung, heart, spleen, liver, kidney and brain.

Example 7

(93) Process Development

(94) Upstream and downstream processes for vaccine manufacture will be established according to standard industry norms in a cGMP grade BSL2 facility (Alvim et. al., 2019). Vero cell lines initially grown in tissue culture flasks at the research scale will be scaled-up in 5 L benchtop, stirred-tank bioreactors (Knowles et. al., 2013). Microcarriers will be used to enhance attachment of the adherent Vero cells. The culture will be maintained in a fed-batch mode with medium replenished on days 3 and 5 with fresh media to boost cellular productivity. VLPs secreted during the cell culture fermentation will be separated from cell mass by tangential-flow filtration (TFF) and downstream purification are undertaken. Concentration and buffer exchange will be conducted using additional TFF steps. This is followed by hydrophobic interaction chromatography and an ion-exchange step to purify the VLP fraction. Purified VLPs will be formulated and sterile filtered as described below (Aravelo et. al., 2016).

(95) Process robustness and reproducibility will be established by running at least 3 lots of VLP production for both USP and DSP steps. Analytical method development includes tests for purity and potency in addition to validation of physical characteristics (appearance, particle size, pH) and identity (ELISA or Western Blot). Purity assessment will be done by SDS-PAGE densitometry as well as total protein content measured by BCA assay. Potency assessment will be conducted by use of validated ELISA methods using antibodies specific to VLP antigens. Impurity testing includes host protein and DNA quantitation.

(96) Vaccines will be tested using a panel of buffers and excipients to identify a stable formulation. Histidine Tris and PBS buffers will be mixed with excipients such as Polysorbate 80, amino acids, sucrose and trehalose. In addition, vaccines will be adsorbed on alum adjuvant for increased

stability (as well as enhanced immunogenicity). Formulated vaccine will be sterile filtered and filled in 2 mL single dose glass vials. Multi dose vials will be tested once vaccine PoC was established. An initial goal is to establish a formulation compatible with a vaccine vial monitor (VMM) 14 label. To do so, formal stability studies will be conducted as described below.

(97) Accelerated stability studies will be conducted at 4° C., 25° C., 37° C. and 40° C. followed by potency assessment allowed selection of the lead formulation from the panel above. For formal stability studies, a minimum of three lots of lead vaccine formulation will be placed on stability study for 3 to 6 months. Stability-indicating assays including pH, physical appearance, potency and sterility will be conducted.

(98) It is to be understood that this invention is not limited to any particular embodiment described herein and may vary. It is also to be understood that the terminology used herein is for the purpose of describing particular embodiments only, and is not intended to be limiting, since the scope of the present invention will be limited only by the appended claims.

(99) Where a range of values is provided, it is understood that each intervening value between the upper and lower limit of that range (to a tenth of the unit of the lower limit) is included in the range and encompassed within the invention, unless the context or description clearly dictates otherwise. In addition, smaller ranges between any two values in the range are encompassed, unless the context or description clearly indicates otherwise.

(100) Unless defined otherwise, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs. Representative illustrative methods and materials are herein described; methods and materials similar or equivalent to those described herein can also be used in the practice or testing of the present invention.

(101) All publications and patents cited in this specification are herein incorporated by reference as if each individual publication or patent were specifically and individually indicated to be incorporated by reference, and are incorporated herein by reference to disclose and describe the methods and/or materials in connection with which the publications are cited. The citation of any publication is for its disclosure prior to the filing date and should not be construed as an admission that the present invention is not entitled to antedate such publication by virtue of prior invention. Further, the dates of publication provided may be different from the actual dates of public availability and may need to be independently confirmed.

(102) It is noted that, as used herein and in the appended claims, the singular forms “a”, “an”, and “the” include plural referents unless the context clearly dictates otherwise. It is further noted that the claims may be drafted to exclude any optional element. As such, this statement is intended to serve as support for the recitation in the claims of such exclusive terminology as “solely,” “only” and the like in connection with the recitation of claim elements, or use of a “negative” limitations, such as “wherein [a particular feature or element] is absent”, or “except for [a particular feature or element]”, or “wherein [a particular feature or element] is not present (included, etc.) . . . ”.

(103) As will be apparent to those of skill in the art upon reading this disclosure, each of the individual embodiments described and illustrated herein has discrete components and features which may be readily separated from or combined with the features of any of the other several embodiments without departing from the scope or spirit of the present invention. Any recited method can be carried out in the order of events recited or in any other order which is logically possible.

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Claims

1. An immunogenic composition comprising a virus-like particle (VLP) and at least one antigenic polypeptide displayed on the surface of the VLP, wherein the at least one antigenic polypeptide is derived from one or more domains of a coronavirus spike(S) protein, a coronavirus membrane (M) protein or combinations thereof, wherein the VLP has at least 95% sequence identity to SEQ ID NO: 167.
 2. A method of inducing an immune response to at least one coronavirus antigen in a subject in need thereof, comprising administering an effective amount of the immunogenic composition of claim 1 to the subject.
 3. The method of claim 2, further comprising the steps of: allowing a suitable period of time to elapse; and administering at least one additional dose of the immunogenic composition of claim 1 to the subject.
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